4 (a) (i)	$R = V^2 / P$ *or* $P = IV$ and $V = IR$	C1	
	$= (220)^2 / 2500$		
	$= 19.4\,\Omega$ (*allow 2 s.f.*)	A1	[2]
(ii)	$R = \rho l / A$	C1	
	$l = [19.4 \times 2.0 \times 10^{-7}] / 1.1 \times 10^{-6}$	C1	
	$= 3.53\,\text{m}$ (*allow 2 s.f.*)	A1	[3]
(b) (i)	$P = 625$, 620 or 630 W	A1	[1]
(ii)	R needs to be reduced	C1	
	Either length ¼ of original length		
	or area 4× greater		
	or diameter 2× greater	A1	[2]
5 (a) (i)	sum of e.m.f.'s = sum of p.d.'s around a loop/circuit	B1	[1]